

SIAM EV Platform

Cargo Space Monitoring API Reference
External Integration Edition

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1 Introduction

This document describes the REST and WebSocket APIs for the **Cargo Space Monitoring** sub-system of the SIAM EV Unified Energy & Logistics Management Platform. It is a scoped extract of the full platform API reference, prepared for handover to an external development partner responsible for the Cargo / Fleet management system. It does not include the OCPP Charging Monitoring or Energy Management System (EMS) APIs, which remain internal to SIAM EV; see Section 10 for what a partner integrator needs to know about those boundaries.

The Cargo Space Monitoring sub-system tracks two things:

- **Trucks** — mobile units tracked by GPS in real time. Each truck carries a load sensor (0–100%) and a camera that photographs the cargo bay automatically when the door closes at departure.
- **Stations** — static named locations (reference points) where trucks arrive and depart. A station has only an identifier, a name, and a geographic position.

Cargo photos are stored in S3-compatible object storage and referenced by time-limited pre-signed URLs.

1.1 Base URL

All endpoints are served under the API Gateway. Replace `{host}` with the deployment address (e.g. `api.siam-ev.example.com`).

```
1 https://{host}/api/v1
```

Listing 1: Base URL pattern

Cargo base path: `/api/v1/cargo`

1.2 Authentication

Every request must carry a Bearer token obtained from the Auth Service.

```
1 Authorization: Bearer <access_token>
```

Listing 2: Authorization header

Header	Required	Description
Authorization	Yes	Bearer <JWT> issued by <code>/auth/token</code>
Content-Type	Yes (POST/PUT)	<code>application/json</code>
X-Request-ID	No	Idempotency key; echoed in response headers

1.3 Common Error Schema

```
1 {
2   "success": false,
3   "error": {
4     "code": "TRUCK_NOT_FOUND",
5     "message": "No truck with id 'truck-bkk-099' exists.",
6     "timestamp": "2026-07-02T10:15:30Z"
7   }
8 }
```

Listing 3: Error response envelope

HTTP Status	Meaning	Common cause
200	OK	Successful read
201	Created	Resource created
400	Bad Request	Missing or invalid fields
401	Unauthorized	Missing / expired token
403	Forbidden	Insufficient role
404	Not Found	Resource does not exist
409	Conflict	Duplicate resource
500	Internal Error	Upstream / database failure

2 Data Models

2.1 Truck Object

```

1 {
2   "id":          "truck-bkk-007",
3   "plateNumber": "GKh-1234",
4   "name":        "Truck 07",
5   "type":        "StraightTruck", // StraightTruck | Trailer | Refrigerated | Van
6   "capacityCbm": 40,
7   "status":      "InTransit",     // Idle | InTransit | Loading | Unloading | Offline
8   "currentLoadPct": 72,           // 0-100; from load sensor in cargo bay
9   "position": {
10    "lat":        13.8562,
11    "lng":        100.6211,
12    "headingDeg": 245,
13    "speedKph":   68,
14    "timestamp":  "2026-07-02T10:04:55Z"
15  },
16  "currentTripId": "trip-20260702-0041",
17  "driverId":      "usr-drv-019",
18  "lastSeenAt":    "2026-07-02T10:04:55Z"
19 }

```

Listing 4: Truck object schema

status	Meaning
Idle	At a station, engine off, no active trip
InTransit	Moving between stations
Loading	Door open, cargo being loaded
Unloading	Door open, cargo being unloaded
Offline	GPS/telemetry signal lost > 5 min

2.2 Route Object

A *route* defines a named sequence of stations a truck is assigned to serve.

```

1 {
2   "id":          "route-bkk-R12",
3   "name":        "Bangkok South Loop",
4   "description": "Lad Krabang Hub -> Samut Prakan -> Bang Na -> Return",
5   "waypoints": [
6     { "order": 1, "stationId": "site-bkk-001",
7       "name": "Lad Krabang Hub", "lat": 13.7284, "lng": 100.7498,
8       "dwellMinutes": 30 },
9     { "order": 2, "stationId": "site-bkk-012",
10      "name": "Samut Prakan Depot", "lat": 13.5990, "lng": 100.5964,
11      "dwellMinutes": 20 },
12     { "order": 3, "stationId": "site-bkk-005",
13      "name": "Bang Na Post Office", "lat": 13.6607, "lng": 100.6050,
14      "dwellMinutes": 15 }
15   ],
16   "estimatedDurationMin": 130,
17   "distanceKm":          68.4,
18   "activeOnWorkdays":   true,
19   "activeOnWeekends":   false
20 }

```

Listing 5: Route object schema

2.3 Trip Object

A *trip* is one end-to-end run of a route by a specific truck. A cargo photo is captured automatically on each `DoorClose` event (i.e. when the door shuts after loading at a station).

```

1 {
2   "id":          "trip-20260702-0041",
3   "truckId":     "truck-bkk-007",
4   "routeId":     "route-bkk-R12",
5   "driverId":    "usr-drv-019",
6   "status":      "InProgress", // Scheduled | InProgress | Completed | Cancelled
7   "isWorkday":   true,
8   "scheduledDepartureAt": "2026-07-02T08:00:00Z",
9   "actualDepartureAt":   "2026-07-02T08:07:00Z",
10  "actualArrivalAt":     null, // null while InProgress
11  "events": [
12    {
13      "eventId":    "evt-tr-0041-001",
14      "type":       "DoorClose", // Arrival | Departure | DoorOpen | DoorClose
15      "stationId":  "site-bkk-001",
16      "stationName": "Lad Krabang Hub",
17      "timestamp":  "2026-07-02T08:05:00Z",
18      "loadPct":    85,
19      "cargoPhoto": {
20        "photoId":    "photo-0041-001",
21        "url":         "https://storage.siam-ev.example.com/...",
22        "urlExpiresAt": "2026-07-02T11:05:00Z", // pre-signed URL TTL
23        "takenAt":    "2026-07-02T08:05:03Z",
24        "fileSizeKb": 342
25      }
26    },
27    {
28      "eventId":    "evt-tr-0041-002",
29      "type":       "Arrival",
30      "stationId":  "site-bkk-012",
31      "stationName": "Samut Prakan Depot",
32      "timestamp":  "2026-07-02T09:44:00Z",
33      "loadPct":    61,
34      "cargoPhoto": null
35    }
36  ]
37 }

```

Listing 6: Trip object schema

2.4 Station Object

A *station* is a named geographic reference point where trucks pick up and drop off cargo. It is static master data: the Cargo sub-system stores only the fields a dispatcher or map view needs to identify and locate the site.

```

1 {
2   "id":          "site-bkk-001",
3   "name":        "Lad Krabang Hub",
4   "address":     "999 Moo 1, Lad Krabang, Bangkok 10520",
5   "lat":         13.7284,
6   "lng":         100.7498,
7   "type":        "Hub" // Hub | Depot | PostOffice | CustomerSite
8 }

```

Listing 7: Station object schema

3 Truck Endpoints

3.1 GET /cargo/trucks — Live truck map data

Returns all trucks with current GPS position and load percentage for the real-time map dashboard.

Query parameters:

Param	Type	Default	Description
status	string	—	Filter by status enum
routeId	string	—	Filter trucks assigned to a route
limit	integer	200	Max records
offset	integer	0	Pagination offset

Response 200:

```
1 {
2   "success": true,
3   "data": [ /* array of Truck objects */ ],
4   "meta": { "total": 18, "inTransit": 11, "idle": 5, "offline": 2 }
5 }
```

Listing 8: Live truck list response

3.2 GET /cargo/trucks/{id} — Get single truck

Returns the full Truck object including the current trip reference.

Response 200:

```
1 { "success": true, "data": { /* Truck object */ } }
```

3.3 GET /cargo/trucks/{id}/trips — Truck trip history

Returns completed and in-progress trips for a specific truck, with optional date and day-type filters.

Query parameters:

Param	Type	Required	Description
from	ISO-8601	No	Filter trips departing after this time
to	ISO-8601	No	Filter trips departing before this time
dayType	string	No	workday weekend all (default all)
routeId	string	No	Limit to a specific route
status	string	No	Completed InProgress Cancelled
limit	integer	No	Max records (default 50)
offset	integer	No	Pagination offset

Response 200:

```
1 {
2   "success": true,
3   "data": [
4     {
5       "id": "trip-20260702-0041",
6       "routeId": "route-bkk-R12",
7       "routeName": "Bangkok South Loop",
8       "status": "InProgress",
9       "isWorkday": true,
10      "actualDepartureAt": "2026-07-02T08:07:00Z",
11      "actualArrivalAt": null,
12      "loadAtDeparturePct": 85,
13      "loadAtArrivalPct": null,

```

```
14     "eventCount":      2
15   }
16 ],
17 "meta": { "total": 143, "limit": 50, "offset": 0 }
18 }
```

Listing 9: Truck trip history response

4 Route Endpoints

4.1 GET /cargo/routes — List routes

Query parameters: activeOnWorkdays (bool), activeOnWeekends (bool).

Response 200:

```
1 { "success": true, "data": [ /* array of Route objects */ ] }
```

4.2 GET /cargo/routes/{id} — Route details

Returns the full Route object including all waypoints with station coordinates.

Response 200:

```
1 { "success": true, "data": { /* Route object */ } }
```

4.3 GET /cargo/routes/{id}/trips — Trips for a route

Returns all trips that ran (or are running) on a given route, across any truck. Supports the same from/to/dayType filters as the truck-level history endpoint.

Response 200:

```
1 {
2   "success": true,
3   "data": [
4     {
5       "id": "trip-20260702-0041",
6       "truckId": "truck-bkk-007",
7       "truckPlate": "gk-1234",
8       "driverId": "usr-drv-019",
9       "status": "InProgress",
10      "isWorkday": true,
11      "actualDepartureAt": "2026-07-02T08:07:00Z",
12      "loadAtDeparturePct": 85
13    }
14  ],
15  "meta": { "total": 92, "limit": 50, "offset": 0 }
16 }
```

Listing 10: Route trip history response

5 Trip Endpoints

5.1 `GET` `/cargo/trips` — Search trips

Global trip search across all trucks and routes with rich filtering — the primary endpoint for the history view and the planning dashboard.

Query parameters:

Param	Type	Required	Description
from	ISO-8601	No	Departure time lower bound
to	ISO-8601	No	Departure time upper bound
dayType	string	No	<code>workday</code> <code>weekend</code> <code>all</code>
truckId	string	No	Scope to one truck
routeId	string	No	Scope to one route
stationId	string	No	Trips that visited this station
status	string	No	Trip status filter
minLoadPct	integer	No	Return trips where departure load $\geq$ value
maxLoadPct	integer	No	Return trips where departure load $\leq$ value
limit	integer	No	Max records (default 50, max 500)
offset	integer	No	Pagination offset

Response 200:

```

1 {
2   "success": true,
3   "data": [ /* array of Trip summary objects (no events array) */,
4   "meta": {
5     "total":      320,
6     "limit":      50,
7     "offset":     0,
8     "avgLoadPct": 71.4,
9     "workdayCount": 234,
10    "weekendCount": 86
11  }
12 }
```

Listing 11: Trip search response (summary list)

5.2 `GET` `/cargo/trips/{id}` — Get full trip detail

Returns the complete Trip object including every event, load reading, and cargo photo URL for that trip.

Response 200:

```

1 { "success": true, "data": { /* full Trip object with events array */ } }
```

Note on cargo photo URLs: Pre-signed URLs embedded in the response expire after 3 hours. Clients should not persist the URL; re-fetch the trip to get a fresh URL if needed.

6 Planning Analytics Endpoints

These endpoints aggregate historical trip data into the metrics needed to plan future route assignments (truck count, trip frequency, load efficiency).

6.1 [GET](#) /cargo/analytics/utilization — Load utilization over time

Returns average load percentage and trip frequency bucketed by time interval and optionally split by route, truck, or day type.

Query parameters:

Param	Type	Required	Description
from	ISO-8601	Yes	Start of analysis window
to	ISO-8601	Yes	End of analysis window
groupBy	string	No	route truck dayType hour
routeId	string	No	Scope to one route
dayType	string	No	workday weekend all
interval	string	No	Bucket size: 1d 1w 1mo (default 1d)

Response 200:

```

1 {
2   "success": true,
3   "data": {
4     "window": { "from": "2026-06-01T00:00:00Z", "to": "2026-07-01T00:00:00Z" },
5     "interval": "1d",
6     "groupBy": "route",
7     "series": [
8       {
9         "groupKey": "route-bkk-R12",
10        "groupLabel": "Bangkok South Loop",
11        "buckets": [
12          {
13            "date": "2026-06-01",
14            "tripsCompleted": 3,
15            "avgLoadPct": 74.2,
16            "peakLoadPct": 95,
17            "minLoadPct": 42,
18            "totalKm": 205.2
19          }
20        ]
21      }
22    ]
23  }
24 }
```

Listing 12: Utilization analytics response

6.2 [GET](#) /cargo/analytics/planning — Fleet planning recommendations

Analyses historical patterns and returns data-driven recommendations for route scheduling, truck count, and travel frequency. This is the primary output for the planning workflow.

Query parameters:

Param	Type	Required	Description
from	ISO-8601	Yes	Historical window start
to	ISO-8601	Yes	Historical window end
routeId	string	No	Limit analysis to one route
targetUtil	number	No	Desired avg load % (default 80)

Response 200:

```

1 {
2   "success": true,
3   "data": {
4     "analysisWindow": { "from": "2026-06-01T00:00:00Z", "to": "2026-07-01T00:00:00Z" },
5     "targetUtilPct": 80,
6     "routes": [
7       {
8         "routeId": "route-bkk-R12",
9         "routeName": "Bangkok South Loop",
10        "workdays": {
11          "avgTripsPerDay": 3.1,
12          "avgLoadPct": 74.2,
13          "peakDemandHour": "08:00",
14          "currentTrucksUsed": 2,
15          "recommendedTrucks": 2,
16          "recommendedFreqPerDay": 3,
17          "utilizationGapPct": 5.8
18        },
19        "weekends": {
20          "avgTripsPerDay": 1.0,
21          "avgLoadPct": 41.3,
22          "peakDemandHour": "10:00",
23          "currentTrucksUsed": 2,
24          "recommendedTrucks": 1,
25          "recommendedFreqPerDay": 1,
26          "utilizationGapPct": 38.7
27        },
28        "insights": [
29          "Weekend demand is 55% lower than workday demand.",
30          "Reallocating 1 truck from weekends could serve 2 additional workday trips.",
31          "Peak load occurs between 07:00 and 09:00 on workdays."
32        ]
33      }
34    ]
35  }
36 }

```

Listing 13: Planning recommendations response

6.3 [GET](#) /cargo/analytics/load-history — Load trend with photos

Returns load readings paired with cargo photo thumbnails for a route or truck over a date range. Designed for the history gallery view in the dashboard.

Query parameters:

Param	Type	Required	Description
from	ISO-8601	Yes	Start date
to	ISO-8601	Yes	End date
routeId	string	No	Filter by route
truckId	string	No	Filter by truck
dayType	string	No	workday weekend all
limit	integer	No	Max events returned (default 100)

Response 200:

```

1 {
2   "success": true,
3   "data": [
4     {
5       "tripId": "trip-20260702-0041",
6       "truckId": "truck-bkk-007",
7       "truckPlate": "gk-1234",

```

```
8      "routeName":      "Bangkok South Loop",
9      "stationName":    "Lad Krabang Hub",
10     "departureAt":     "2026-07-02T08:07:00Z",
11     "isWorkday":        true,
12     "loadPct":          85,
13     "photo": {
14       "photoId":        "photo-0041-001",
15       "url":             "https://storage.siam-ev.example.com/...",
16       "thumbnailUrl":    "https://storage.siam-ev.example.com/.../thumb",
17       "urlExpiresAt":    "2026-07-02T11:07:00Z",
18       "takenAt":         "2026-07-02T08:05:03Z"
19     }
20   },
21 ],
22 "meta": { "total": 187, "limit": 100, "offset": 0 }
23 }
```

Listing 14: Load history with photos response

7 Station Endpoints

Stations are static master data. These endpoints populate map markers, filter dropdowns, and the route builder. They carry no real-time energy or environmental data.

7.1 `GET` /cargo/stations — List stations

Query parameters: type, limit, offset.

Response 200:

```
1 {
2   "success": true,
3   "data": [ /* array of Station objects */ ],
4   "meta": { "total": 18 }
5 }
```

7.2 `GET` /cargo/stations/{id} — Get station

Returns the Station object (id, name, address, lat, lng, type).

8 Alert Endpoints

8.1 `GET` /cargo/alerts — List active alerts

Query parameters: stationId, truckId, severity (Critical | Warning), limit, offset.

Response 200:

```
1 {
2   "success": true,
3   "data": [
4     {
5       "alertId":      "alrt-20260702-007",
6       "sourceType":   "Zone",           // Zone | Truck
7       "sourceId":     "zone-stn-001-cold-01",
8       "type":         "TemperatureExceeded",
9       "severity":     "Critical",
10      "value":         9.1,
11      "threshold":     8.0,
12      "unit":          "degC",
13      "triggeredAt":   "2026-07-02T10:22:00Z",
14      "acknowledgedAt": null
15    }
16  ]
17 }
```

Listing 15: Active alerts response

Note on sourceType: Truck alerts originate from sensors on the vehicle (e.g. a refrigerated compartment). Zone alerts originate from station-level environmental monitoring that is configured and owned by an internal SIAM EV system outside this document's scope — treat `sourceId` as an opaque string for display/filtering purposes.

8.2 `POST` /cargo/alerts/{alertId}/acknowledge — Acknowledge alert

Request body:

```
1 { "acknowledgedBy": "usr-0007", "note": "Compressor maintenance dispatched." }
```

9 WebSocket — Live Cargo and Fleet Feed

A single WebSocket connection delivers both fleet (truck) and station (zone) events. Use the `scope` parameter to subscribe only to the events your dashboard panel needs.

```
1 wss://{host}/api/v1/cargo/ws?scope=trucks,zones&token=<access_token>
```

Truck position and load update:

```
1 {
2   "event": "TruckPositionUpdate",
3   "truckId": "truck-bkk-007",
4   "data": {
5     "lat": 13.8562,
6     "lng": 100.6211,
7     "headingDeg": 245,
8     "speedKph": 68,
9     "loadPct": 72,
10    "timestamp": "2026-07-02T10:04:55Z"
11  }
12 }
```

Listing 16: TruckPositionUpdate event

Cargo photo ready event (emitted seconds after door closes at a station):

```
1 {
2   "event": "CargoPhotoReady",
3   "tripId": "trip-20260702-0041",
4   "truckId": "truck-bkk-007",
5   "data": {
6     "eventId": "evt-tr-0041-001",
7     "stationId": "site-bkk-001",
8     "stationName": "Lad Krabang Hub",
9     "loadPct": 85,
10    "photoId": "photo-0041-001",
11    "thumbnailUrl": "https://storage.siam-ev.example.com/.../thumb",
12    "takenAt": "2026-07-02T08:05:03Z"
13  }
14 }
```

Listing 17: CargoPhotoReady event

All WebSocket event types:

event	scope	Trigger
TruckPositionUpdate	trucks	GPS reading every 5 s while InTransit
TruckStatusChange	trucks	Status transitions (e.g. InTransit → Idle)
TripStarted	trucks	New trip begins (truck departs station)
TripCompleted	trucks	Trip ends at final waypoint
CargoPhotoReady	trucks	Photo captured on DoorClose; thumbnail available
EnvironmentUpdate	zones	Zone sensor reading every 30 s
AlertTriggered	zones	Threshold breach on zone or truck sensor
AlertResolved	zones	Reading returns within threshold

10 Integration Notes

This section summarizes, at a high level, how the Cargo system fits into the wider SIAM EV platform. Details of the other sub-systems are intentionally excluded from this handover document.

- **Station identity is shared internally.** A `stationId` (e.g. `site-bkk-001`) may correspond to the same physical site in other internal SIAM EV systems. This cross-reference is coordinated entirely on the SIAM EV side; the Cargo API contract in this document does not change based on it, and no additional integration work is required by the Cargo system to support it.
- **Zone-sourced alerts.** As noted in Section 6, some entries returned by `GET /cargo/alerts` originate from station environmental monitoring that is owned and configured internally. The Cargo system only needs to display and allow acknowledgement of these alerts via the documented endpoints — it does not need to compute or validate them.
- **Demand-response and energy coordination.** SIAM EV runs internal workflows that may temporarily prioritize certain stations (e.g. during a facility power event). These do not require any Cargo API changes or calls to systems outside this document.
- **Out of scope for this handover.** OCPP charging telemetry, the Energy Management System (EMS) API, and the outbound webhook registry are separate SIAM EV-internal systems and are not included here. If a future feature requires the Cargo system to call one of those APIs directly, SIAM EV will provide a supplementary interface specification at that time.

11 API Quick Reference

Endpoint	Method	Description
<i>Fleet</i>		
/cargo/trucks	GET	Live truck positions and loads (map)
/cargo/trucks/{id}	GET	Single truck detail
/cargo/trucks/{id}/trips	GET	Trip history for a truck
/cargo/routes	GET	List routes
/cargo/routes/{id}	GET	Route detail with waypoints
/cargo/routes/{id}/trips	GET	All trips on a route
/cargo/trips	GET	Global trip search with date/dayType filter
/cargo/trips/{id}	GET	Full trip detail with events and photos
<i>Analytics</i>		
/cargo/analytics/utilization	GET	Load % and trip frequency over time
/cargo/analytics/planning	GET	Fleet sizing and frequency recommendations
/cargo/analytics/load-history	GET	Load trend with cargo photo gallery
<i>Stations (static reference data)</i>		
/cargo/stations	GET	List stations (name, position, type)
/cargo/stations/{id}	GET	Single station
/cargo/alerts	GET	Active alerts (trucks and stations)
/cargo/alerts/{alertId}/acknowledge	POST	Acknowledge an alert
/cargo/ws	WS	Live truck and station event feed